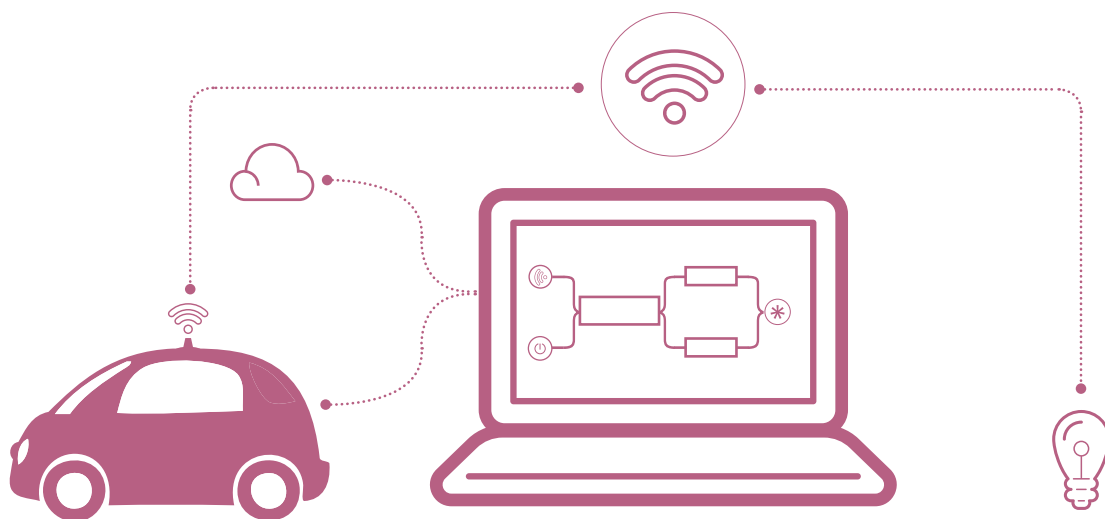


Enhancing the Design of Autonomous Cars

Using Node-RED

Autonomous cars are the future. A lot of work goes into designing them to minimise the risk of machine failure and accidents. This article outlines how Node-RED can be used to create flows for developing animations of car movements.



Autonomous cars are set to revolutionise the travel experience for all of us. As per a Gartner report, there were 508,622 autonomous cars on the road in 2021. This figure is expected to go up to 745,705 in 2023. Autonomous cars reduce the possibility of accidents caused by driver distraction, reduce traffic congestion, and optimise the use of fuel. Cameras, radars, and ultrasonic and touch sensors used in these cars replace human decision-making. This article outlines how to enhance the capability of these cars by creating Node-RED flows that develop animations for car movements. The flows are implemented and the

movements are observed with various input and output sequences, using the animations developed.

Node-RED

Node-RED is open source software used for the Internet of Things (IoT) and for making Web applications. Unlike normal software that use complex programs and structures, Node-RED uses GUI operations to create 'flows' that are easy to understand. This is done by integrating various 'nodes' without using any complex code. Similar to the successful serverless architecture, the zero coding techniques of Node-RED are gaining popularity for IoT

operations. Node-RED allows us to develop real-time applications in IoT, online services and projects with minimal effort. This article will help you create a pleasant graphical representation of blocks named 'nodes' in Node-RED, which provide programmers a fantastic way to develop applications. It can also function as a wiring tool that can connect a large number of both hardware and software devices.

Node-RED development environments can also connect hardware devices to Web services. Users can share their flows with each other, making group based activities convenient. Node-RED can be used